
PREDICTING ROAD ACCIDENT RATE IN FLORIDA: APPLICATION OF MULTIPLE LINEAR REGRESSION, HIERARCHICAL LINEAR MODELS AND KRIGING

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Abstract

This project focuses on identifying models/methods that can be used for predicting road accidents rate on locations with recorded and unrecorded data. The models/methods used are developed using limited data with very few explanatory variables. The techniques used are illustrated using accident rates over 56 years from 1960-2016 for each month of the year, in 51 traffic analysis zones around Florida. Making accurate predictions with limited data is extremely important as low income countries that suffer from the highest accident rates and have limited access in data and resources need to form targeted strategies to improve road safety. So the aim of this project is to find models and methods that can work with limited explanatory variables and still be accurate. The method followed here is to first fit multiple regression models with a number of possible subsets of the explanatory variables and then selecting the one with the smallest mean square error (MSE) which was obtained through 10 fold cross validation. After that four different hierarchical models are fit and their mean square errors, obtained through 10 fold cross validation, are compared with the multiple regression models. The last method used is kriging in order to make predictions and identify more or less dangerous locations in areas where we do not have data available. This method is evaluated using the mean square prediction error. Results indicated that the multiple regression model which included all three explanatory variables and an interaction term had a lower MSE when compared to the hierarchical linear models, and as such was preferred. It was found that kriging generated accurate predictions when using the mean rate at each location as the predictor variable. In general it was found that multiple regression and kriging are appropriate methods that can be used to make predictions both in sampled and unsampled locations. Multiple linear regression and kriging have been used before. However in most cases multiple linear regression has been used when there are plenty of explanatory variables. Kriging is a method that has been more widely used in other fields, but is not a very common method in the field of road statistics, and as such is further investigated in this project.

Keywords Accident rate prediction · Multiple linear regression · Hierarchical linear models · Spatiotemporal modelling · Seasonality

1 Introduction

1.1 Background

Road safety is of great importance, as it is predicted that by 2030, road traffic injuries will be the fifth leading cause of death, accounting for 3.6% of total deaths. This percentage stood at 2.5% in 2018, where road traffic injuries occupied the eighth place on the leading causes of deaths worldwide [17]. World Health Organisation (WHO) estimates that more than 1.35 million road users die each year and estimates that 20-50 million road users suffer non-fatal injuries. However here it should be mentioned that death rate of road accidents has stabilized and declined from 135 deaths for every 100,000 vehicles in 2000 to approximately 64 deaths for every 100,000 vehicles in 2016, in relation to the increasing world population and vehicles [17]. This represents more than a 50% decline during these years [17]. However this decline is not uniform across all countries. From 2013 - 2016 no reductions of road deaths were recorded in low income countries, followed by some reductions in 23 middle income and more reductions in 25 high income countries [17]. This highlights the inequalities across regions and the need to develop methods that poorer countries can use to prevent accidents. Economic and social costs of road accidents stress even more the importance of road safety. These costs include expenses such as material damage, medical costs, law enforcement and legal costs, as well as loss of time. These costs are borne both by the taxpayers and the individuals who are directly involved in an accident [6]. It is estimated that these costs account for approximately 1 and 3% of a nation's GDP [8], with the percentage being lower in low- and middle-income countries regardless the fact that they have higher fatality rates. The inequality in road safety among countries can be understood even better from the fact that low income countries have an average death rate of 27.5 per 100,000 which is almost triple when compared to the death rate in high income countries (8.3 per 100,000). What is even more surprising is the fact that 13% of road deaths occur in low income countries despite the fact that only 1% of motor vehicles are present in these countries. Not only rates are much higher in low income countries, laws surrounding road safety are more lenient and do not meet up the good practice standard. Drink and speed laws are indicative as only 2% and 13% of low income countries follow best practice laws. The statistics are similar when it comes to vehicle safety, helmet use and safe roads laws [17]. So it is more than clear, that low income countries are in more need of improving road safety.

There are several measures that can be adopted to prevent accidents. A successful example is the Netherlands where there has been a significant decrease in road accidents from 2005 to 2020. This has happened partly due to a learning society, but also infrastructural adaptation, safer vehicles and more safety legislation and enforcement [25]. These are measures that can be implemented by most countries. However, the cost of implementation is an issue, especially for low and middle income countries. For this reason identifying accidents hotspots and making accurate predictions of accidents rate is very important, so that countries can follow more targeted strategies, and the inequality in road safety between low and high income countries can be improved.

Accidents occur for different reasons and there are many factors that can affect accident rates. In a lot of relevant studies, there are sufficient data that can be used when modelling accident rates. Typical explanatory variables include time (year, month, day), location, vehicles involved, type of road, as well as other factors like drivers age and gender, weather conditions, speed of vehicle, involvement of alcohol etc [15], [1], [22]. However in some cases there is limited explanatory variables, but it is still very important to model accident rates and identify hotspots in the future. This is one of the main issues in low income countries where accident rates are very high but data access and collection is very limited. So it is vital that new methods need to be developed that will work well with less predictor variables, as this will be a step towards improving road safety in poorer countries where resources are limited. So, this research focuses on investigating whether accidents rates can be modeled accurately with very few explanatory variables.

1.2 Aim, objectives and structure of the paper

The aim of the project is to model collision rates by using just the three explanatory variables that are present in the available data set. These are the year, the month and the zone. These three explanatory variables are recorded in 51 traffic analysis zones around Florida. More specifically, this paper aims to investigate whether using multivariate linear regression is an appropriate method for modeling accident rates. This will be done by fitting various possible models and comparing their mean squared errors, calculated through 10-fold cross validation, for evaluation purposes. In addition, the paper investigates whether there are any appropriate spatial methods which can be used in order to fit a surface over the region of interest and make predictions of collision rates on locations where data is not recorded. These findings will be represented through useful

visual graphs. Finally, the paper will investigate whether hierarchical models can be used in order to get a better idea of the variation in accidents rates between the different locations/months.

The paper is divided into 6 Sections. Section 1.3 describes relevant related work which was taken into account and acted as a guide when progressing with the project. Section 2 describes the models and methods used throughout the project. This section provides a more theoretical background in order to support the reader's understanding around these topics. Section 3, describes the data and the application of the methodologies on the data. Section 4 provides a detailed explanation of the results generated from Section 3, and Section 5 identifies the main conclusions as well as areas of future work. Section 6 includes the appendices and at the end someone can find the references.

1.3 Related work

1.3.1 Multiple regression and generalised linear models

Multiple regression has been a wide known approach for modelling and predicting accidents rates. Alqatawna et.al [2], in a study that investigated multiple linear regression models for predicting highway traffic accidents in Spain, found that multiple regression can give very accurate results. The study used data of accident counts in Spain freeways, over 4 years between 2014 to 2016 and was based on 6 explanatory variables including speed, segment length, slope gradient and radius of horizontal curve in each section, heavy vehicle percentage and annual average daily traffic. Results showed an R squared value of 0.903 indicating a good performance model. Similarly a study in Thailand about forecasting road traffic deaths using data over 20 years (1997-2020), found that multiple linear regression was among the most accurate models investigated. This study was based in 5 explanatory variables, which included economic and energy consumption factors such as the GDP and the energy consumption of the transport section of each year, as well as the number of registered motorcycles, cars and trucks for every year. To select the appropriate variables the study used 4 techniques. Firstly, the models was fitted with all variables, and then forward selection, backwards elimination and stepwise selection were all tested. The adjusted coefficient of determination as well as the mean absolute percentage error (MAPE) was used for evaluation purposes. The three final models selected had model accuracies between 84 and 88%. The three models were tested against time series (ARIMA) models, curve estimation models and path analysis models. The study concluded that the model with the lowest MAPE (6.4%) was one of the regression models tested. ARIMA models were not appropriate for the Thailand data set [9]. However, a study in Nigeria in 2016 that used time series ARIMA models to forecast accident rates and data from 1960 - 2013 showed that there were significant variations in the accident rates trends over the years [23]. As such time is included as a predictor in the project. It was clear from the studies above that multiple regression is a powerful method that can lead to very accurate results. For this reason it was chosen as one of the main focuses of this project.

A second approach that has been widely used is generalised linear models, where outcomes do not need to be normally distributed. It is very common in road accidents analysis to deal with count data or with skewed distributions for accident rates. Some of the most commonly used models include poisson and negative binomial models, which possess most of the desirable statistical properties that are not present in multiple linear regression [15]. It has been argued that poisson and negative binomial regression can be more appropriate since they allow for different variance from the mean [11]. However these models need count data, which is not the case in this project and hence can not be used here.

1.3.2 Hierarchical/multilevel linear models

Hierarchical models have also been employed for the study of road accidents, when there is correlated data. An example of data correlation can be that of vehicles with specific characteristics which may lead to better survival probabilities in an accident. Hence the casualties in the vehicle may have more similar outcomes [10],[14]. A study in 2003 in Norway of road accidents casualties between 1985-1996 showed significant residual variation in casualty outcomes between accidents and different locations. The study aimed to investigate whether these variations can be explained by risk factors for which the police has available information from past records. These risk factors included casualties and vehicle characteristics (age, sex, type of vehicle, drinking, seat belt usage) as well as the accidents circumstances (weather, season, road type, etc). There were three levels in the analysis. Level 1 represented the hierarchy of accidents, level 2 represented within accidents variation and level 3 within municipality variations. This analysis was based on 18 explanatory variables and found presence of correlation between the same or accidents or accidents in the same municipality suggesting that the hierarchical model approach can give better estimate models when compared to more traditional approaches [10]. Similarly multilevel models have been used to model crash severity. Poisson multilevel

models were used in a study in 49 locations in Greece between 1998 to 2002, concerning the impact of the intensification of law enforcement in road accidents in different locations. The results indicated that intensified police enforcement, had a significant impact on both accidents reduction as well as fatalities and there was significant regional variation effect. From this study it is evident that multilevel models can have a clear benefit [29]. Hence in this project a hierarchical structure is included for each zone, and also extended for each month in order to investigate if variation is presents there.

1.3.3 Spatial models

Spatial models have also been used for road accidents analysis as it has been shown that accidents occurring in nearby locations can be correlated, which needs to be accounted in a model. Such models assume that explanatory variables coefficients vary across locations [21]. A study in Seoul that used spatial lag (SLM) and error models (SEM) to analyse traffic crashes in Seoul, showed that the SEM had the best goodness of fit followed by the SLM and last came the normal regression model. The models were assessed based on the AIC and the models were fitted with 33 explanatory variables. The results of the study indicate that spatial autocorrelation needs to be accounted for as it can provide more accurate results [21]. So the spatial autocorrelation was chosen to be investigated. However the data used for the project were not appropriat for these methods, so other techniques were investigated.

Another common approach for identifying hotspots and making predictions for areas where accidents have not occurred is kernel density estimation (KDE) as well as kriging. The later is much less investigated in the context of road accidents. The main assumption in these methods and in geostatistics in general is the existence of spatial autocorrelation which needs to be investigated before proceeding with using these techniques. These two methods were compared in a study in 2015, which investigated hotspot identification in Minnesota using data between 2003 and 2007. The data included information on the outcome of the accident as well as the time of the day and the conditions under which they occurred. The two methods were assessed using the prediction accuracy index (PAI) where the higher the number the better the performance. The results indicated that kriging has a better performance when compared to the traditional KDE. As such the method used in this project is kriging. Moreover the two methods generated slightly different results when it comes to the hotspots identified. In general the study concludes that kriging seems a powerful technique which needs further investigation in the field of road safety [24].

2 Methodology

2.1 Multiple Linear Regression

Multiple linear regression was applied in order to investigate whether there is a linear relationship between the accidents rates (response variable) and the three explanatory variables which include year, month and location. It was used as a method to understand how the year, zones and months affect accident rate and how they can be used for making future predictions. Multiple linear regression assumes that the relationship between the response and explanatory variables is linear. Further assumptions include that multicollinearity between explanatory variable is not present, the residuals are independent with constant variance and are normally distributed. The general formula of linear regression is as follows:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i \quad (1)$$

In the equation above, y_i represents the response/dependent variable, $x_{i1}, \dots, x_{i2}$ represent the explanatory/independent variables, β_0 is the constant y intercept, $\beta_1, \dots, \beta_p$ are the regression coefficients representing the change in y with respect to X_i given that all other values remain constant, and ϵ_i represent the residuals. The model selection approach followed in this project is a best subset selection approach, where a number possible combinations of variables have been fitted. For assessing the models the mean square error (MSE) is used which is obtained after performing 10 cross validation to the model, in order to get a better estimate of its predictive ability. This procedure is followed for every model fitted.

2.2 Hierarchical models

Hierarchical linear models (HLM) are very similar to ordinary least squares regression and follow the same assumptions except the one of the independence of errors. This assumption is not met as HLM allow for data

across levels to be correlated. Explanatory variables are separated into fixed and random, where random effects are categorical variables allowing for information sharing between levels. In this project HLMs are used in order to investigate whether there are significant variations in the zones or months effects, and if yes whether the models produce accurate predictions. The general formula is as follows:

$$Y_{ij} = \beta_{0j} + \beta_{1j} + X_{ij} + r_{ij} \quad (2)$$

Here, Y_{ij} is the response for individual i under level j , X_{ij} is the predictor variable for individual i nested under level 2 unit j , r_{ij} is random error associated with the i_{th} level-1 unit nested within the j_{th} level-2 unit, β_0 is the intercept for the j_{th} level-2 unit, and β_{1j} is the regression coefficient associated with X_{ij} for the j_{th} level-2 unit [28].

2.3 Kriging

Kriging is a geostatistical method of spatial interpolation that uses a limited set of data to make predictions of a variable over a continuous surface. In this paper kriging is used as a method to identify locations with higher or lower accidents rate on a 2d space. Moreover kriging is used as a visual tool in order to identify how accidents rate changes in space. Kriging is based on models that take into account autocorrelation between the known data points and assumes that the Euclidian distance or direction between data points reflects a spatial correlation that can be used to explain variation in the surface. More specifically, the way kriging works, is that the predicted values are weighted average of known data, and the weights are determined in a way that they are unique to each predicted point and a function of the separation distance (lag) between the observed location and the location to be predicted [24]. Kriging requires some assumptions to be satisfied. These are that the data are spatially autocorrelated and that they are normally distributed. There are different types of kriging available. One of the most widely used, is ordinary kriging. Ordinary kriging assumes that the mean and variance are constant across the spatial field. The general formula for ordinary kriging is as follows:

$$\hat{z}(\mathbf{x}_0) = \sum_{i=1}^N \lambda_i z(\mathbf{x}_i) \quad (3)$$

z is the kriging function used to make the prediction. $\hat{z}(\mathbf{x}_0)$ denotes the value that we want to predict in a specific location x_0 and acts as a dependent variable, $z(\mathbf{x}_i)$ is the known value at location i acting as a regressive variable, λ_i is the unknown weight for the measured value at location i and N is the number of known observations. In order to calculate λ the expected prediction error is minimized subject to the constrain that all λ_i sum to 1. This can be accomplished by taking the derivatives of the Lagrangian with respect to λ . Finally to perform kriging an empirical variogram is fitted and then an appropriate model for the data is selected. The variogram is used to show the variability between data points as a function of distance and can be calculated as follows:

$$2\gamma(h) = \frac{1}{N(h)} \times \sum_{n=1}^{N(h)} [z_n - z_{n+h}]^2 \quad (4)$$

where N_h is the number of data pairs at distance h inside the searching area, z_n is the value at location n and z_{n+h} is the value at location $n + h$. The results of this method are discussed in detail in Section 4.

3 Data and model application

3.1 Data

The data used in the project consist of accident collision rates over 56 years from 1960-2016 for each month of the year, in 51 traffic analysis zones around Florida. These are presented in Figure 1. The original data contained a lot of missing values and were organised in a way that did not facilitate the analysis and did not include the year column. To make analysis easier data were organised in four columns containing the

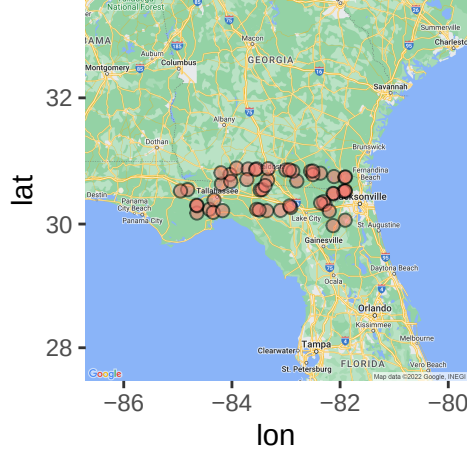


Figure 1: Map with the 51 zones around Florida

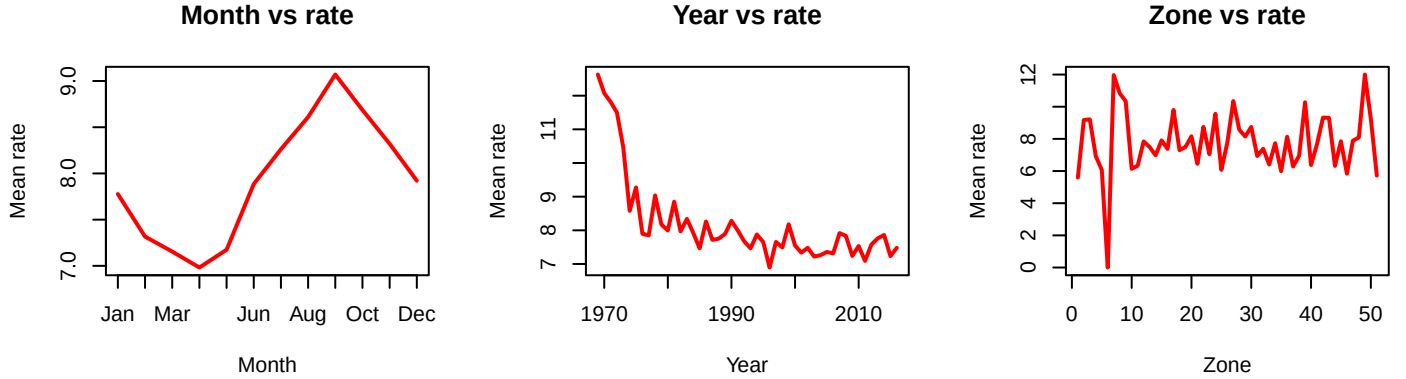


Figure 2: Exploratory analysis of data

rate (response variable), the year, the month and the zone (explanatory variables). The data then consisted of 34221 rows with 24715 containing missing (NA) values. After all missing values were removed the first year with data was 1969 and the data consisted of 9506. The first few rows of the dataset before and after preprocessing can be found in Appendix 6.1. The packages used for the data manipulation were dplyr [26] and tidyr [27].

Figure 2 shows some graphical summaries of the data. The results indicate that there could be specific zones and seasons that suffer from higher accident rates. More specifically the mean rate starts increasing in April, where the mean rate is the lowest and peaks in September. After that mean starts declining gradually. In general it seems like the spring months show lower rates (<7.2) when compared to winter and autumn months which show rates ranging from 7.3 to 7.9 and 8.1 to 9.1 respectively. Moreover it is clear that as the years change from 1969-2016, the mean rate decreases significantly until around 1990. After that the decline is slower, with 2000 and 2001 being exceptions with very low mean rates. Finally, it seems that zones also experience some fluctuation in mean rates, with zone 6 having rate of 0. On the other hand some zones (7,8,9,24,27,39,49) experience higher rates in general ranging from 9.6 in zone 24 to 12 in zone 49. After investigating more zone 6 it was found that there are only 14 records of rated for this specific zone, and all of them are from the year 2000 with just one being from 2021. As such zone 6 data are removed for the rest of the analysis as the data are not enough to produce accurate results.

3.2 Model application

3.2.1 Multiple linear regression

In section 2.1 it is mentioned that for this project a best subset selection approach is followed and the best model is selected based on the mean square error (MSE) value after performing 10 fold cross validation. Before fitting any of the models a transformation is required. Before the transformation, rate seems to follow a Log-normal distribution. Thus, a log transformation ($\log(\text{rate} + 0.001)$) is applied (see Appendix 7.1). The addition of 0.001 is required as there are zeroes in the data which would lead to undefined values. The general equation for the models is:

$$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone} + \epsilon_i \quad (5)$$

where *year* is the year that the rate was recorded, *month* is the month that the rate was recorded and *zone* is the location where the rate was recorded. It should be mentioned that the zone variable is treated as an indicator variable and not as continue variable. The models in table 1 were fitted and cross validation was performed in order to get the MSE. Not all possible subsets are fitted as the model with all two way interactions as well as the model with the three way interaction had higher MSEs than simpler models and the number of parameters was higher than the number of observations leading to less accurate predictions. As such overly complicated models were not fitted. The estimates of the regression coefficients in each models were obtained via maximum likelihood, and then cross validation was performed using the caret [12] package in R . The results indicating the best fitting models and the significant explanatory variables are presented in section 4.

Model	Equation
Model 1	$y_i = \beta_0$
Model 2	$y_i = \beta_0 + \beta_1 * \text{year}$
Model 3	$y_i = \beta_0 + \beta_1 * \text{zone}$
Model 4	$y_i = \beta_0 + \beta_1 * \text{month}$
Model 5	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{zone}$
Model 6	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month}$
Model 7	$y_i = \beta_0 + \beta_1 * \text{month} + \beta_2 * \text{zone}$
Model 8	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone}$
Model 9	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone} + \beta_4 * \text{year} * \text{month}$
Model 10	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone} + \beta_4 * \text{zone} * \text{month}$
Model 11	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone} + \beta_4 * \text{year} * \text{zone}$
Model 12	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone} + \beta_4 * \text{year} * \text{month} * \text{zone}$
Model 13	$y_i = \beta_0 + \beta_1 * \text{year} + \beta_2 * \text{month} + \beta_3 * \text{zone} + \beta_4 * \text{year} * \text{zone} + \beta_5 * \text{year} * \text{month} + \beta_6 * \text{month} * \text{zone}$

Table 1: Models fitted for multiple linear regression

3.2.2 Hierarchical linear models

In section 2.3 it was mentioned that a technique which we will follow is that of the hierarchical/mixed linear (HLM) effects models. It was mentioned that the idea behind the approach is that HLM include having fixed

Table 2: All hierarchical models fitted

Model	1	2	3	4
Fixed.effects	None	None	Year and Month	Year and Zone
Random.effects	Zone	Month	Zone	Month

and random effects, with random effects varying withing groups and allowing for information sharing between levels. In this case groups include the different months and different zones. For this study four different models are fitted using the lmer [3] and lmerTest [13] packages in R.

The first two models are the two random intercept models that include the month and the zone respectively. The two other models are mixed models which include the year and either the month or the zones as a fixed effects. The models are summarized in table 3. The models fitted are evaluated after performing 10 fold cross validation and obtaining the mean square error (MSE). The cvms [16] package was used for cross validation. The MSE was used in order to compare the models with the ones obtained in section 3.2.1. The results are discussed and compared in more detail in section 4.

3.2.3 Kriging

Applying kriging to the data was more complex as the data were not in the right format. The main packages used for kriging application were rgdal [4], gstat [19], sp [20] and sf [18]. All of these packages offer wide functionality in the field of geostatistics. As mentioned in section 2.3. kriging is used in order to make predictions and identify more or less dangerous locations in areas where we do not have data available.

In order to make these predictions two separate approaches were followed. The first approach included taking the regression coefficients of the zones of the model selected in section 3.2.1 and using them as the prediction variable. Then the coefficients are joined with the longitude and latitude values for each zone. After that a sample/empirical variogram is fitted which is used in order to identify if there is spatial correlation in the data and how it changes over distance. If the correlation is present the variogram helps identify any appropriate model that can be used for kriging. After having found an appropriate model, the dataset from above, is then converted into a spatial object, in order for the data to be distinguished from coordinates, which can be used for kriging. Other than that a grid is created which will be used in order to perform kriging to the unsampled locations. The grid of points represent locations where rate will be predicted by using the chosen fitted model in order to visually represent how the rate changes in space. The second approach included taking the mean rate in each zone as the prediction variable instead of the zone coefficients and then following the same steps as before to fit the variogram. The second approach was followed because the results from the first approach indicated that spatial correlation was not present there. The approaches and the findings are compared and discussed in detail in section 4.2. Ten fold cross validation, using the gstat package [19], is used to assess the predictive performance of kriging. The way it works is that each time a prediction for a sampled location is used, using the other known values and the prediction is compared to the observed value.

4 Results and discussion

4.1 Multiple linear regression results

As mentioned in section 2.1 the approach followed for this project was a best subset selection approach where all possible models were fitted and the one with the lowest MSE was selected. Table 3 shows all the models fitted along with their corresponding MSE. The full models can be found in table 1 in section 3.2.1. The MSE was obtained through 10 cross validation which was performed using the caret [12] package.

Table 3: Mean square errors (MSE) for all models fitted

Model	1	2	3	4	5	6	7	8	9	10	11	12	13
MSE	0.1230	0.1147	0.0877	0.1142	0.0871	0.1082	0.0814	0.0800	0.0796	0.0917	0.0734	0.2231	0.0904

From table 3 it is clear that model 11 is the one with the lower MSE, followed by model 9. Model 1 which includes just the intercept has the highest MSE as expected. In general it seems that the MSEs are low

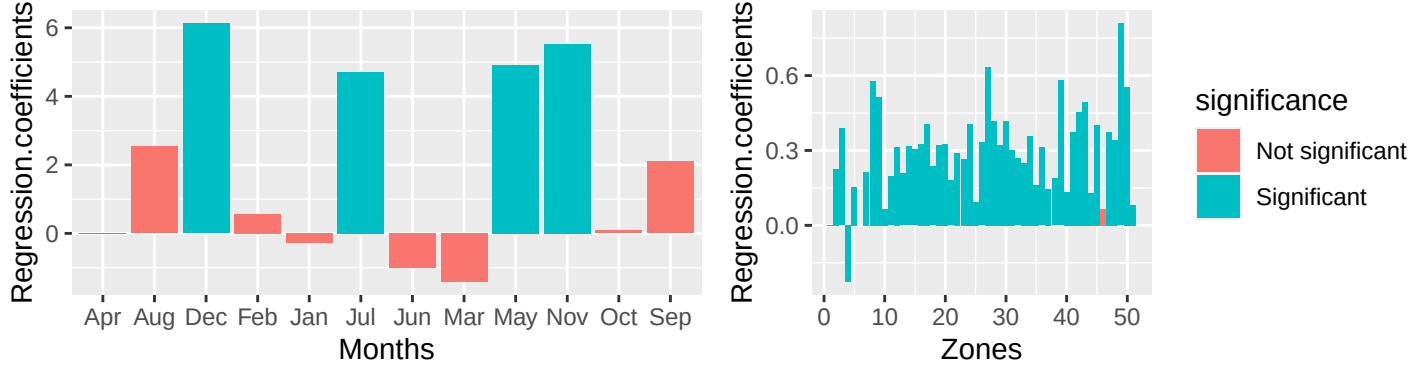


Figure 3: Plot of significant regression coefficients for months (left) and zones (right)

considering that the log rate ranges from -6.91 to 3.42, which indicates that multiple linear regression is appropriate for this specific dataset.

Looking at all the MSEs of models 2,3 and 4 it is clear that model 3 which includes just the zones variable has the lowest MSE. Moreover comparing models 5,6 and 7 it is again clear that models 5 and 7 which include the zone as one of the two explanatory variables have lower MSEs when compared to model 6 which does not include the zone variable. From these two observations it is evident that adding the zones variable provides benefit to the model, indicating that the rates vary significantly by zone. On the other hand models 12 and 13 which include the three way interaction and all the two ways interactions respectively, have higher MSEs indicating that over complicating the models do not lead to a model with better predictive ability. In addition models with a lot of interactions are more difficult to interpret. The models with the lowest MSEs are models 9 and 11 which include the zone and month interactions with the year. Comparing with model 10 that includes the interaction of zone with month and has higher MSE it is clear that adding an interaction that includes the year improves the predictive ability of the model. After much consideration the preferred model is model 9 which includes all the variables along with the interaction between year and month. This model is the one with the second lowest MSE. The reason for choosing this model instead of the one with the lowest MSE is the fact that the number of model parameters for all the other models with the two way interactions as well as the model with the three way interaction is greater than the number of observations in the dataset making the models less applicable to smaller datasets. Since the project focuses on finding ways that can work well with limited data and can benefit low income countries which experience this problem, model 9 is selected. The regression coefficients of the selected model are presented in Figure 3 and in full in Appendix 7.2.

After performing a simple anova analysis it is found that all components are significant with p-values less than 0.001. Here it should be noted that the reference level for zones and months is zone 1 and April respectively. From Appendix 7.2 we observe that year has a negative coefficient indicating that increasing years result in lower accident rates. This is not surprising as from figure 1 there was a declining trend for rates from 1960 to 2016. From Figure 3 (right), most zones have positive coefficients and are significantly associated with the rate indicating an increase in rate when compared to zone 1. We also observe that zone 4 has a negative regression coefficient indicating a decrease in rate when compared to zone 1. The only zone which does not have a statistically significant p-value is zone 46. From Figure 3 (left) it seems that only four (November, December, July, May) are statistically significant. This could indicate an effect of the different seasons as the significant months represent each season of the year. All of their regression coefficients are positive indicating an increase in rate during these months when compared to April. The most positive increase is for December, followed by November, May and July. The fact that the regression coefficients for the other months are not statistically significant means that the level of risk for these months are more static in terms of their level of risk. Moving on, the interaction term tells us whether the month difference in accident rate differs based on the year. The fact that the interaction term is significant means that difference in months accident rate varies depending on the year, or in other words changes in time, and should be included in the model. The interactions that are significantly associated with rate include the months that were significant above. The regression coefficients of the significant interactions are all very small negative indicating that as year increases the difference in rate between these months and April is becoming smaller. This means

Table 4: MSEs of the four HLMs fitted in section 3.2.3

Models	1	2	3	4
MSE	0.1129	0.1384	0.1077	0.1077

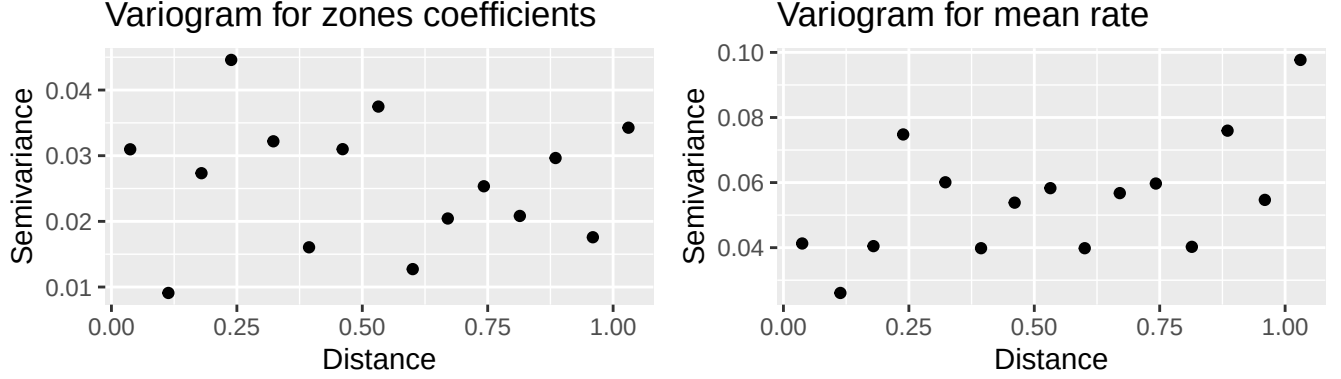


Figure 4: Sample variograms: Variogram when using the zone coefficients (left) and variogram when using the mean rate (right)

that either these months are becoming safer or that April is becoming more dangerous. These interactions could be an indication of climate changing during the years and changing weather conditions having different effects on accident rates. The non significant interactions indicate that for these months their effect on rate does not changes significantly over time, and their level of risk is kind of stable during the years.

4.2 Hierarchical linear/mixed effects models

As mentioned in section 3.2.2 the approach followed here was fitting the four models, comparing their MSEs and the comparing with the models from the simple linear regression. The MSEs obtained for the models in table 2 are summarised in table 4.

From table 4 it is clear that the models which include the fixed effects have lower MSE when compared to the models that include only the random intercepts. Model 3 and Model 4 seem to have the same MSE, but in fact they are just very similar (0.107677 for Model 3, 0.107679 for Model 4). However, comparing table 3 with table 4 it is evident that most models from table 3 have lower MSEs. Moreover the selected model has still the lowest MSE indicating that the multiple linear regression is more appropriate for this specific dataset. This means that probably the effects that are treated as random (Months and Zones) are quite distinct and are not influenced from each other. The sharing of information between the random effects is not necessary in this case. This is confirmed after calculating the intra-class correlation for zones and months which was 0.212 and 0.055 respectively. Moreover, the fact that there is sufficient data for each month/zone create more certainty over what is happening in each level and information sharing is not necessary. However the MSE for the HLM models is still quite low which indicates that they still perform well.

4.3 Kriging

In Section 3.2.3 it is mentioned that one of the first steps in order to use kriging is to fit a sample variogram to investigate spatial autocorrelation and fit an appropriate model. However the assumption of normality also needed to be checked. To do that the Shapiro-Wilk test for normality was used, which rejects the hypothesis of normality if the p-value is less than 0.05. It was found that the p-value was 0.5066 when using the zone coefficients as a predictor and 0.0213 when using the mean rate as a predictor. This meant that mean rate needed a transformation. The distribution of mean rate was positively skewed so a square root transformation was applied (Appendinx 7.3.). After the transformation the p-value was 0.0920 and as such the assumption of normality was satisfied. The variaograms from the two approaches mentioned in Section 3.2.3 are presented in Figure 4.

From Figure 4 we can observe that there when using the zone coefficients there does not appear to exist any spatial pattern. It seems that there is a random scatter on the plot. On the other hand, looking at the variogram for mean rate there seems to be an upwards trend and in general points seem to have some kind of pattern. To further explore this observation a Moran I test is conducted. The Moran I test, measures spatial autocorrelation based on both feature locations and feature values at the same time. Under the Moran I statistic the null hypothesis states that the attribute investigated is randomly distributed among the features in the study area. The test gives back a p-value. The p-values for the test conducted were 0.189 when using the zone coefficients and $2.46 \cdot 10^{-7}$ when using the mean rate. This confirms that when using the mean rate there is spatial autocorrelation, while for zones coefficient there is not. This means that the effect of collision rate does not change in any systemic way as we move around the space. On the other hand, taking the mean rate and not separating out the effect of year and month/season, then there is a spatial pattern. This links directly to the result from Section 4.1 where it was found that the preferred model included the interaction term between year and month, which indicated that the accident rate does not change equally over time and month. So both results suggest that there is a spatial structure/pattern in the effect of season or the effect of year.

After having identified the spatial patterns and made sure that the required assumptions are satisfied, we fit a model to the sample variogram. After investigation of the variogram an Exponential model seems appropriate. The parameters of the model, include the nugget (y-intercept), the range which is the distance at which the model first flattens out, and the sill which is the value at which the model first flattens out. After examining Figure 4 (right plot) the initial values of the parameters for the model in the `fit.variogram` function in `r` were chosen. Nugget was 0.04, range was 100 and sill was 0.07. The model was then fitted to the data results are presented in Figure 5. Figure 5 shows that the exponential model fits the data quite well. After that ordinary kriging was performed and results are shown in Figure 5 (right). The points on Figure 5 represent the locations where data were recorded.

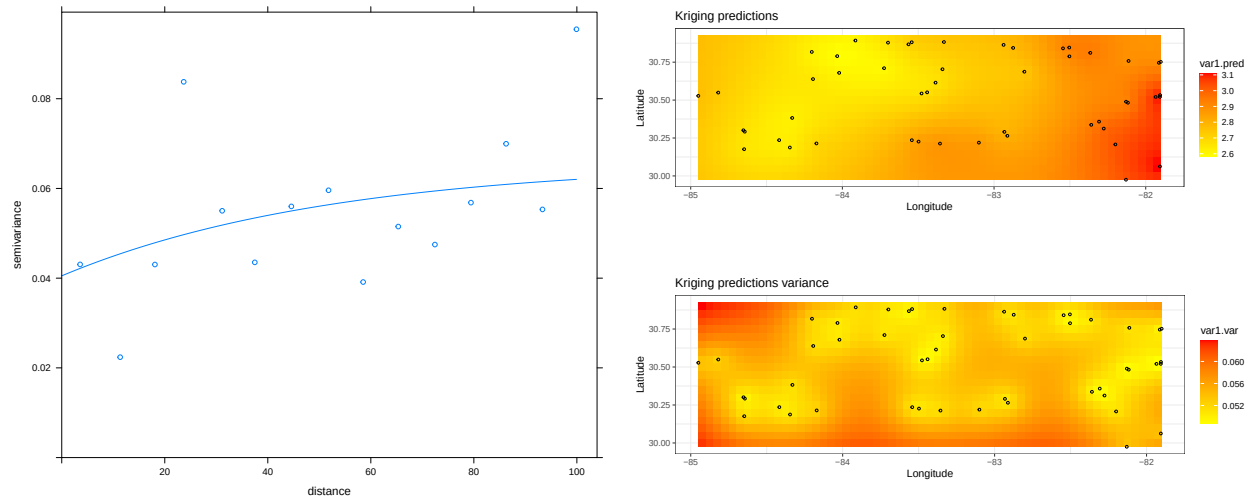


Figure 5: Exponential model fit to the sample variogram of mean rate (left) and kriging plots (right): Kriging prediction (top) and prediction variance (bottom)

Linking Figure 5 with Figure 1 someone can see how kriging has made predictions (Figure 5:top) for the unsampled locations. The rate is higher for locations around Jacksonville. This is not surprising as it is most populated city in the state, which probably means that there is more traffic and hence more accidents. In general this appears to be the pattern as in areas closest to cities (Valdosta, Lake City, Tallahassee) the predicted rate seems to be higher when compared to the center of the grid where there are no big cities. The square root accident rate varies from 2.6 - 3.1. This can easily be converted to the rate by taking the square of each prediction. Moreover the predictions variance (Figure 5:bottom) seems to be quite low which makes the prediction more accurate.

Table 5: Statistics from 10 cross validation on kriging

Statistics	Mean squared prediction error (MSPE)	Correlation (observed vs predicted)	Correlation (predicted vs residual)
Values	0.04699	0.61725	-0.06813

In section 3.2.3 it is mentioned that kriging was assessed using 10 fold cross validation. The results are presented in table 5. Table 5 shows that the mean squared prediction error (MSPE) is 0.047 indicating a good predictive ability of kriging. Similarly the correlations between observed and predicted/residuals indicate that the model is appropriate for predictions. In general kriging is an appropriate method for making prediction for unsampled locations as long as the data exhibit spatial autocorrelation and satisfy the normality assumption.

5 Conclusions and future study

5.1 Conclusions

This project shows a way to overcome the problem of having limited data for making future accident rate predictions for both sampled and unsampled locations. It was found that multiple linear regression gave accurate predictions, when it included all available explanatory variables and an interaction term between year and months. The MSE for the chosen model, calculated through 10 fold cross validation, was 0.0796. The results of the chosen model found that as year increases rate decreases. Moreover it was found that December, November, May and July are significant with positive coefficients indicating an increase in rate during these months when compared to April. Most the the zones were significantly associated with rate, with positive coefficients indicating that they are more dangerous compared to zone 1. Finally the interaction term being significant means that difference in months accident rate varies in time. The significant interactions were the ones including the significant months. Their respective regression coefficients were small negative indicating that as the years go by, these months become safer or April, the reference level, becomes more dangerous.

Hierarchical models were also considered as both the zone and month variables include levels. The models were evaluated using the MSE, calculated through 10 fold cross validation. The results indicated that the preferred multiple regression model had a better prediction ability. The intra-class correlation within the month and zone levels also indicated that levels are not needed for these specific data.

The project also investigates if kriging is an appropriate prediction method in the field of road statistics. It was found that when using the zone coefficients, of the chosen regression model, as the predictor variable, the data did not exhibit any spatial autocorrelation, indicating that accident rate does not change in any systemic way as we move from zone to zone. Since there was not any spatial autocorrelation kriging could not be performed. To further investigate if kriging applies to the data, the mean rate was taken as the predictor variable and then spatial autocorrelation was detected. This meant that the year and month effects lead to some kind of spatial pattern. The results of kriging indicated that kriging is appropriate as the mean squared prediction error, found through 10 fold cross validation, is 0.04699. The predictions heat map, indicated that the rate is predicted to be higher in locations closer to bigger cities. This make sense as bigger cities experience more traffic. The predictions were the highest in the area around Jacksonville which is the biggest city in the state of Florida.

This project showed a way of how countries, with limited data and resources can use their data to make predictions for the future in locations where data exist, but also for locations with unrecorded data.

5.2 Future study

Further work could include investigating whether a Bayesian approach, which is not investigated in the project, is appropriate for limited predictor variables and could generate a model with better predicting ability. The fully bayes approach has been gaining ground the past few years as it has been argued that empirical approaches, used in the past, present disadvantages both on the results but also on the input needed to implement this approach [7], [5]. Moreover, it is mentioned in section 1.3.3 that kernel density estimation (KDE) has been a method with wider application in the field of road statistics when compared to kriging. Just like kriging, KDE is used to estimate a variable in unsampled locations. As such, KDE could be employed for this project and the results could be compared with the ones generated by kriging, in order to identify the most suited method.

6 Appendices

6.1 Appendix

Table 6: First 15 rows of the data before preprocessing

V1	Zones	Rate
9	V2	NA
9	V3	NA
9	V4	NA
9	V5	NA
9	V6	NA
9	V7	NA
9	V8	NA
9	V9	NA
9	V10	NA
9	V11	NA
9	V12	NA
9	V13	NA
9	V14	NA
9	V15	NA
9	V16	NA

Table 7: First 15 rows of the data after preprocessing

	year	zones	MonthAbb
5511	1969	3	Sep
5515	1969	7	Sep
5521	1969	13	Sep
5525	1969	17	Sep
5526	1969	18	Sep
5530	1969	22	Sep
5532	1969	24	Sep
5538	1969	30	Sep
5546	1969	38	Sep
5556	1969	48	Sep
5562	1969	3	Oct
5566	1969	7	Oct
5572	1969	13	Oct
5576	1969	17	Oct
5577	1969	18	Oct

6.2 Appendix

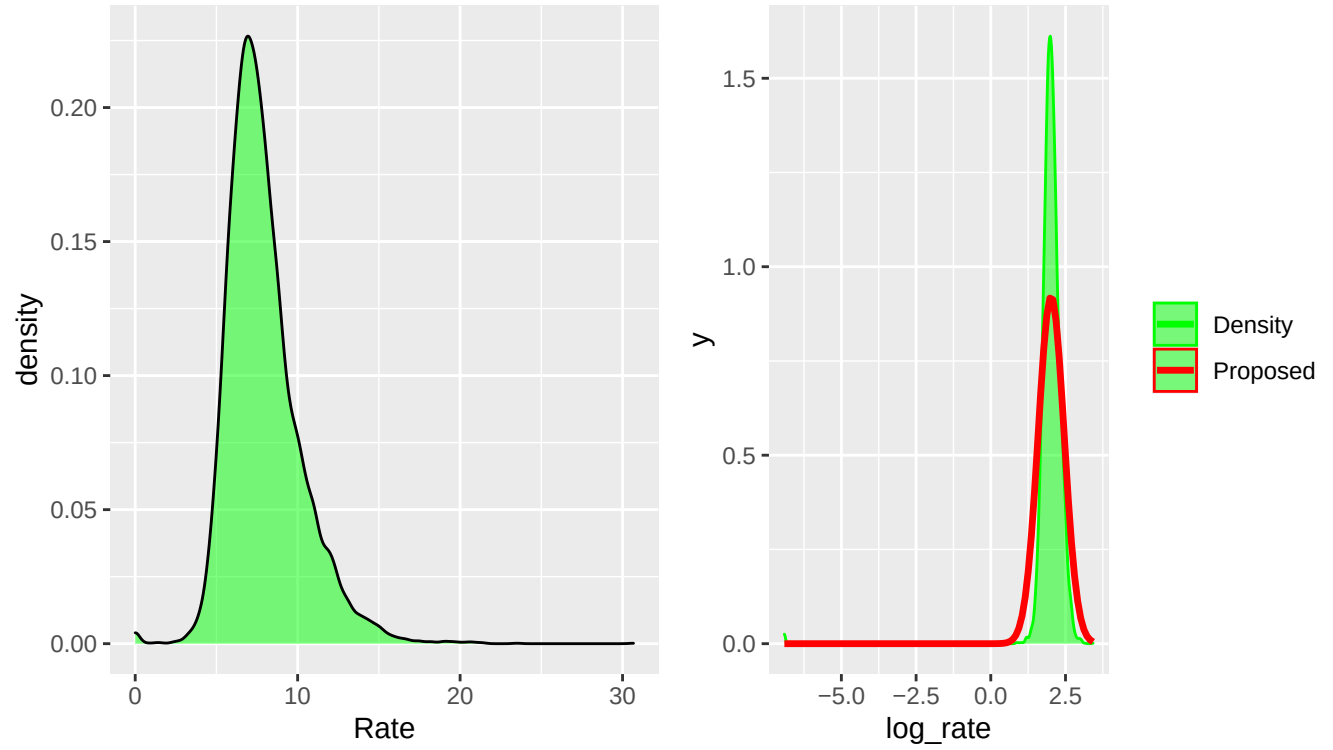


Figure 6: Figure showing the rate distribution before and after transformation as well as the proposed normal distribution”

6.3 Appendix

Table 8: Regression coefficients for interaction term model

term	estimate	std.error	statistic	p.value
(Intercept)	10.0537578	1.9163322	5.2463544	0.0000002
year	-0.0042447	0.0009598	-4.4226224	0.0000099
factor(zones)2	0.2252964	0.0523451	4.3040556	0.0000169
factor(zones)3	0.3878066	0.0384605	10.0832309	0.0000000
factor(zones)4	-0.2269401	0.0454203	-4.9964474	0.0000006
factor(zones)5	0.1530425	0.0398323	3.8421715	0.0001228
factor(zones)7	0.2129246	0.0501702	4.2440461	0.0000222
factor(zones)8	0.5783449	0.0459822	12.5775947	0.0000000
factor(zones)9	0.5131066	0.0386818	13.2647920	0.0000000
factor(zones)10	0.0666867	0.0330470	2.0179360	0.0436262
factor(zones)11	0.1958662	0.0518837	3.7750983	0.0001609
factor(zones)12	0.3149738	0.0468483	6.7232664	0.0000000
factor(zones)13	0.2077617	0.0344656	6.0280936	0.0000000
factor(zones)14	0.3163764	0.0553021	5.7208712	0.0000000
factor(zones)15	0.3042818	0.0380946	7.9875367	0.0000000
factor(zones)16	0.3260760	0.0363642	8.9669502	0.0000000
factor(zones)17	0.4063259	0.0463509	8.7663047	0.0000000
factor(zones)18	0.2362964	0.0323428	7.3059995	0.0000000
factor(zones)19	0.3205832	0.0487604	6.5746633	0.0000000
factor(zones)20	0.3261269	0.0433223	7.5279161	0.0000000
factor(zones)21	0.1799392	0.0356676	5.0448988	0.0000005
factor(zones)22	0.2885941	0.0389674	7.4060309	0.0000000
factor(zones)23	0.2650163	0.0352280	7.5228800	0.0000000
factor(zones)24	0.4068918	0.0412544	9.8629867	0.0000000
factor(zones)25	0.0944235	0.0378770	2.4928971	0.0126876
factor(zones)26	0.3318660	0.0774479	4.2850210	0.0000185
factor(zones)27	0.6350055	0.0340515	18.6483777	0.0000000
factor(zones)28	0.4179611	0.0542466	7.7048378	0.0000000
factor(zones)29	0.3206136	0.0523645	6.1227264	0.0000000
factor(zones)30	0.4176844	0.0323428	12.9142958	0.0000000
factor(zones)31	0.3007547	0.1405876	2.1392680	0.0324396
factor(zones)32	0.2679169	0.0326831	8.1974062	0.0000000
factor(zones)33	0.2497554	0.0809813	3.0841118	0.0020475
factor(zones)34	0.3587071	0.0351671	10.2000817	0.0000000
factor(zones)35	0.1607659	0.0792903	2.0275615	0.0426331
factor(zones)36	0.3115752	0.0648179	4.8069324	0.0000016
factor(zones)37	0.1459592	0.0454231	3.2133256	0.0013165
factor(zones)38	0.1873140	0.0324318	5.7756192	0.0000000
factor(zones)39	0.5798016	0.0566632	10.2324227	0.0000000
factor(zones)40	0.1348042	0.0421677	3.1968602	0.0013939
factor(zones)41	0.3743698	0.0375434	9.9716464	0.0000000
factor(zones)42	0.4546849	0.0408667	11.1260365	0.0000000
factor(zones)43	0.4948281	0.0328082	15.0824476	0.0000000
factor(zones)44	0.1304638	0.0444415	2.9356317	0.0033367
factor(zones)45	0.4012382	0.0382812	10.4813457	0.0000000
factor(zones)46	0.0654284	0.0448535	1.4587134	0.1446773
factor(zones)47	0.3746194	0.0494167	7.5808215	0.0000000
factor(zones)48	0.3407745	0.0326765	10.4287330	0.0000000
factor(zones)49	0.8085318	0.0367391	22.0073890	0.0000000
factor(zones)50	0.5550137	0.0369775	15.0094886	0.0000000
factor(zones)51	0.0809265	0.0383259	2.1115338	0.0347527
MonthAbbAug	2.5327313	2.6485351	0.9562763	0.3389572
MonthAbbDec	6.1246493	2.6065862	2.3496822	0.0188100

term	estimate	std.error	statistic	p.value
MonthAbbFeb	0.5658327	2.6112624	0.2166893	0.8284552
MonthAbbJan	-0.2809450	2.6073052	-0.1077530	0.9141939
MonthAbbJul	4.7205659	2.6073538	1.8104816	0.0702530
MonthAbbJun	-1.0188417	2.6081694	-0.3906348	0.6960760
MonthAbbMar	-1.4179004	2.6116595	-0.5429117	0.5872035
MonthAbbMay	4.9072370	2.6048029	1.8839187	0.0596067
MonthAbbNov	5.5360267	2.6089485	2.1219379	0.0338690
MonthAbbOct	0.0809475	2.6128805	0.0309802	0.9752860
MonthAbbSep	2.0977878	2.6118808	0.8031714	0.4218960
year:MonthAbbAug	-0.0011664	0.0013270	-0.8789403	0.3794561
year:MonthAbbDec	-0.0030032	0.0013062	-2.2992155	0.0215145
year:MonthAbbFeb	-0.0002478	0.0013081	-0.1894628	0.8497342
year:MonthAbbJan	0.0002084	0.0013062	0.1595271	0.8732571
year:MonthAbbJul	-0.0022796	0.0013062	-1.7452247	0.0809784
year:MonthAbbJun	0.0005767	0.0013066	0.4413875	0.6589426
year:MonthAbbMar	0.0007294	0.0013084	0.5574606	0.5772260
year:MonthAbbMay	-0.0024436	0.0013049	-1.8726037	0.0611541
year:MonthAbbNov	-0.0026779	0.0013073	-2.0483531	0.0405531
year:MonthAbbOct	0.0000679	0.0013093	0.0518534	0.9586466
year:MonthAbbSep	-0.0009201	0.0013088	-0.7030355	0.4820510

6.4 Appendix

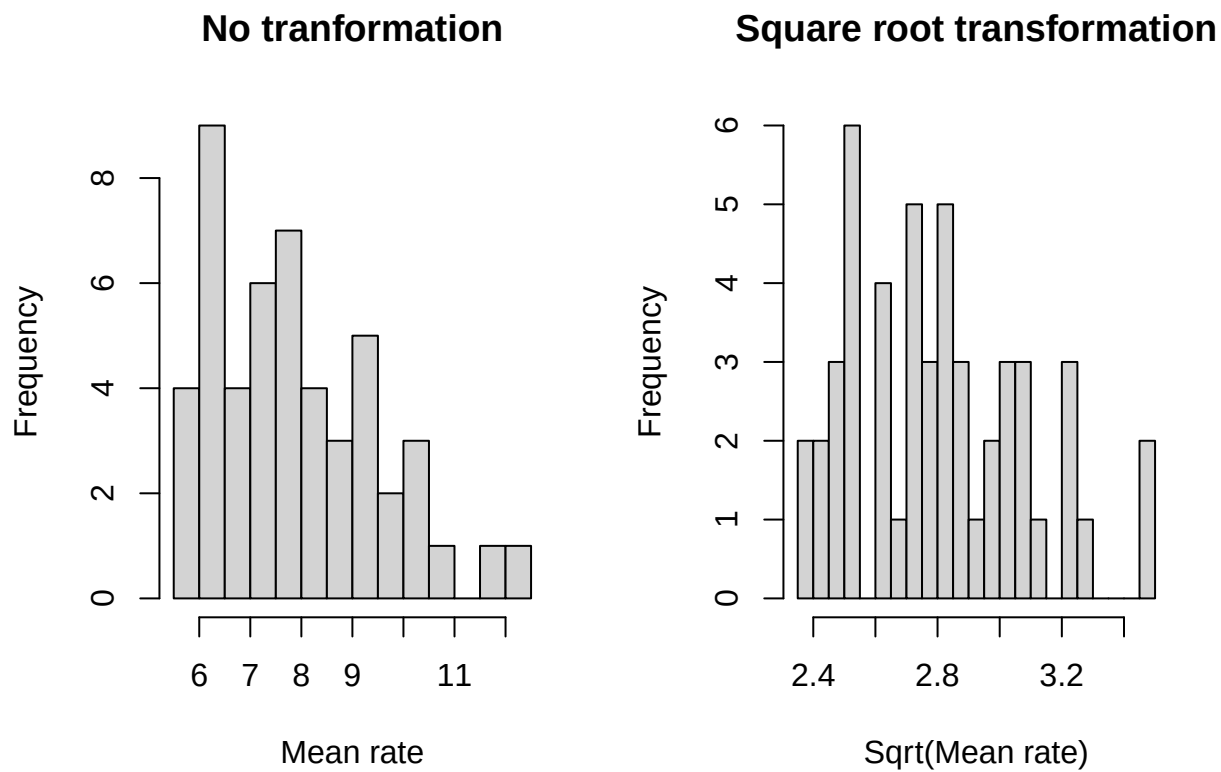


Figure 7: Histograms of mean rate before (left) and after square root transformation (right)

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